Algebra 1 – Module 4 Test PRACTICE

Full Name: _____

Module 4 — Linear Functions: Slope and Slope-Intercept Form Period: _____ _____ / 30

Answer each question, making sure to show your work in the space provided. Take the time to check your work. Write neatly. Circle your final answer.

1. Vocabulary. Write the letter of the term that matches each description. One term is never used. A) x-intercept B) slope C) y-intercept D) rise E) slope-intercept form (4)

a. _____ the vertical change between two points

b. _____ the point where a line crosses the y-axis

c. _____ the ratio of rise to run; how steep a line is

d. _____ an equation written as y = mx + b

2. Find the slope of the line through (2, −2) and (4, 0). (2)

m = _____

3. Find the slope of the line through (0, 6) and (−8, 0). Simplify. (2)

m = _____

4. Find each slope. (a) through (−3, −1) and (−1, −1) (b) through (5, 6) and (5, −6) (2)

(a) _____ (b) _____

5. Name the slope m and the y-intercept b of each line. (a) y = −3 − 2x (b) y = 8 (2)

(a) m = ____ b = ____ (b) m = ____ b = ____

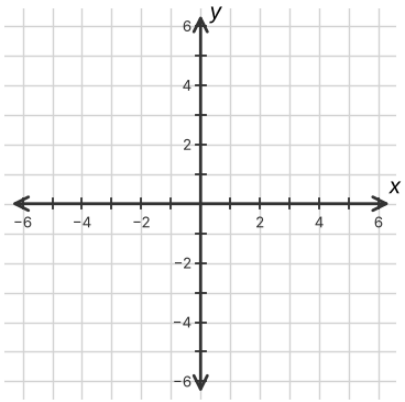
6. Write in slope-intercept form: 2x + y = 3 (2)

y = _____

7. Write in slope-intercept form: 5x − 4y = 8 (2)

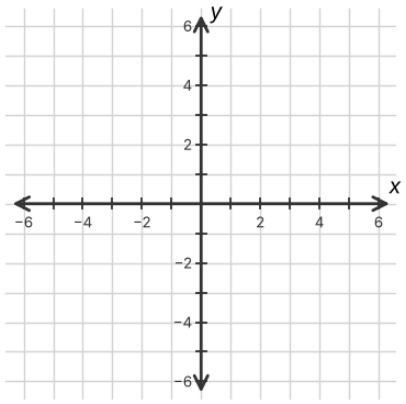
y = _____

8. Graph: y = −(1/4)x − 1 (2)



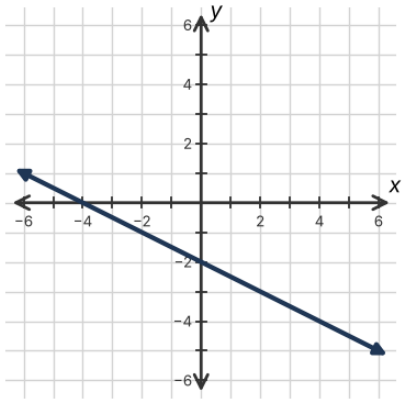
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9. Write in slope-intercept form, then graph: $x + y = -3$ (2)



y = _____

10. Write the equation of the line in slope-intercept form. (2)



y = _____

11. A pool drains at a steady rate. After 2 hours it holds 520 gallons. After 5 hours it holds 400 gallons. Find the slope and the y-intercept, tell what each means, and write the equation (x = hours, y = gallons). (2)

y = _____

12. Error analysis. Marco rewrote $3x - 4y = 8$ as $y = (3/4)x + 2$. Name the mistake in one sentence, then write the correct equation. (2)

correct: y = _____

13. A line passes through (0, 6) and (2, 2). Write its equation in slope-intercept form. (2)

y = _____

Cumulative Review — ACT Style. Circle the best answer. No work required, but scratch work never hurts.

14. Solve: $3x - 3 = x + 3$ (A) 0 (B) −3 (C) 5 (D) 3 (1)

15. $f(x) = -4x - 6$. Find $f(-4)$. (F) −22 (G) 2 (H) 10 (J) −14 (1)